

Rayane Bencharef

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EDUCATION

École de Technologie Supérieure de Montréal (ÉTS) <i>Master of Science (M.Sc.) in Automated Manufacturing Engineering</i> - Thesis Track: Specialization in Artificial Intelligence & Computer Vision. - Awarded "Excellent" mention for Master's Thesis and named to the ÉTS Dean's Honour List (Fall 2025). - Jury recommendation for the Best Master's Thesis Award (Excellence Award).	09/2023 - 12/2025 Montréal, QC
ISIS Castres (INSA partner) <i>Master of Engineering (M.Eng.) in Software Engineering (CTI-accredited degree)</i>	09/2019 - 12/2025 Castres, France
European University of Cyprus <i>Student Exchange in Software Engineering (Erasmus)</i>	02/2023 - 06/2023 Nicosia, Cyprus

PEER-REVIEWED PUBLICATION

International Conference on Computer Vision (ICCV), VisionDocs Workshop <i>Spotlight / Best Paper Award</i> <i>DIVE-Doc: Downscaling foundational Image Visual Encoder into hierarchical architecture for DocVQA.</i> Code Repository Model Weights	10/2025 Honolulu, Hawaii
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RESEARCH EXPERIENCE

Research Assistant in Deep Learning - Multimodality & Interpretability <i>ÉTS Montréal, Synchromedia</i> - Investigating the mechanistic interpretability of Vision-Language Models to uncover how they encode 2D spatial and semantic layouts for DocVQA. - Mentoring a graduate intern in designing the LLM orchestration and retrieval architecture for a RAG-based telecommunication network analysis system.	02/2026 - Present Montréal, QC
Graduate Researcher - Computer Vision & Model Optimization <i>ÉTS Montréal, Synchromedia</i> - Reduced the computational cost of a Large Vision-Language Model in DocVQA by studying two distillation approaches between heterogeneous architectures, halving latency (896ms → 446ms).* - Fine-tuned the GEMMA LLM decoder with a hierarchical visual encoder for DocVQA using QLoRA, improving performance from 80.20 to 82.67 ANLS.* - Investigated positional encoding in Vision Transformers (ViT) using 2D Fourier features, increasing performance from 83 to 84 ANLS. - Studied how VQA models handle structure and layout understanding through document classification and layout analysis tasks (interpretability).* - Adapted RAG pipeline to upgrade single-page Document Understanding VLM to process multi-page documents without adding parameters for industrial applications. *First Author: These works culminated in the DIVE-Doc paper, receiving the Best Paper Award at the ICCV 2025 VisionDocs workshop.	01/2024 - 11/2025 Montréal, QC

INDUSTRY EXPERIENCE

Intern Data Scientist <i>Atout Majeur Concept</i> Engineered an automated predictive pipeline and trained SVM models on limited clinical data to forecast hospital stay durations (78% accuracy).	06/2023 - 08/2023 Toulouse, France
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Freelance Data Engineer / Analyst

09/2022 - 12/2022

TrainPreddict / Linkypharm.fr

Remote

- Architected scalable data models for telemetry datasets and engineered geographic statistical visualizations for healthcare distribution patterns.

Intern Data Scientist in Time Series

05/2022 - 08/2022

CHU Toulouse

Toulouse, France

- Conducted exploratory feature analysis on emergency medical records and built ARIMA/LSTM forecasting models to predict temporal call volumes (80% accuracy).

Software Developer

07/2020 - 08/2022

Horus HealthCare Systems / TrainPreddict

Castres, France

- Designed and deployed robust full-stack backend architectures (Django, Node.js) for healthcare tracking (INCA) and sports telemetry systems.

SELECTED PROJECT

C.O.R.A. (Pediatric Companion Robot)

03/2026 – Present

Systems Architecture, Edge AI, Human-Robot Interaction (HRI), Multimodal Perception

Montréal, QC

- Leading the conceptual design and global system architecture for an interactive companion robot dedicated to pediatric support in hospital environments.
- Collaborating with researchers in pediatrics and neuroscience to refine the system architecture, and conducted an observational site visit at CHU Sainte-Justine to align workflows with clinical constraints.
- Evaluating hardware specifications and memory-efficient multimodal perception modules to ensure safe human-robot interaction (HRI) for mobile agents.

Mila Venture Startup Challenge Fellow

02/2026

Technology Transfer, Systems Architecture

Montréal, QC

- Intensive one-month challenge focused on bridging the gap between advanced AI engineering and viable market applications.

DIVE-Doc Platform

12/2025

Python, FastAPI, Docker, Gradio

- Developed and deployed an interactive web interface (FastAPI/Gradio) to demonstrate the LVLm inference capabilities from the ICCV 2025 publication. [Deployed Demo](#) | [GitHub Repository](#)

Cyprus Fish Species Recognition App

04/2023

Python, ConvNext, ONNX, MLFlow, Docker

- Fine-tuned a ConvNext vision model on a custom dataset (95% accuracy) and optimized the architecture using ONNX for low-latency inference. [Deployed Demo](#) | [GitHub Repository](#)

SKILLS

Research Areas & Machine Learning: Computer Vision (Perception, Classification, Segmentation), Explainable AI (XAI) & Mechanistic Interpretability, Model Optimization (Distillation, Quantization, Edge Inference), Multimodal Foundation Models (VLMs, VQA), Time Series Forecasting

HPC & Research Engineering: HPC Workload Manager (Slurm), Distributed Training, Experiment Tracking (MLflow), Containerization (Docker), Automated Testing (Pytest), Cloud & Infrastructure (Azure, Terraform, FastAPI)

Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face (accelerate, datasets, transformers), ONNX/ONNX Runtime, OpenCV, Scikit-learn

Programming Languages: Python, SQL, R, Java, JavaScript

Developer Tools: CUDA, Git/GitHub, Visual Studio Code

Languages: French (Native), English (Professional)